P1 Report — Factors Affecting Final Grades (UCI Student Performance, `mat`)

Summary version. Full report (data, code, charts, tests): `report.pdf`.

Main question: How do study time, absences, family support and intermediate grades (G1, G2) relate to the final grade (G3)?

1. Executive summary

Question	Answer (mathematics, 357 students with G3 > 0 unless stated)
Intermediate grades → G3	Strongest link: r(G2, G3) = **0.966**, r(G1, G3) = 0.892
Past failures → G3	0 failures: **11.94**, 1: 10.15, 2: 8.83, 3: 8.27 (r = -0.294)
Absences → G3	low 12.26 / medium 11.38 / high 10.37 (r = **-0.213**, ANOVA Holm p < 0.001)
Study time → G3	weak positive, r = 0.127 (ANOVA Holm p 0.019, eta^2 = 0.036: small)
Internet at home → G3	yes 11.68 vs no 10.71; raw p 0.027, Holm p 0.080 → **not significant after correction**
All factors together (regression)	background factors explain R^2 = 0.234; significant in both courses with the others held fixed: **studytime (+1 bucket), failures (+1 past failure), absences (+1 absence), schoolsup (yes vs no)**; internet adjusted b = +0.53 (p 0.247)
Family support (`famsup`) → G3	yes 11.35 vs no 11.8 (Holm p 0.404): **no significant difference**
Pass rate	67.1% of all 395 students; 74.2% after removing 38 dropout suspects
Early-warning rule (no G3)	designed on mat: recall 93.8%, precision 81.9%; **tested on unseen Portuguese-only students: recall 91.4%, precision 66.0%**

Key data-quality finding: 38 students have G3 = 0 and **38 of them have 0 recorded absences**. They are treated as *dropout suspects* (flagged, not deleted). Including them flips the absences correlation to r = 0.034 and hides the real negative effect.

2. Limitations and next steps

- One subject/year as primary cohort; small groups (see Chart 1 `n=` labels) give wide confidence intervals.
- Regression adjusts only for variables in the data; unmeasured factors (motivation, teachers, income) can still confound, so effects are associations, not causes.
- "Dropout suspect" is inferred from the data, not confirmed by the schools; the sensitivity analysis (6.1) shows the absences conclusion does not depend on it.
- The holdout set shares the schools and the school year with the training data, and the strict holdout is small (n = 275); precision on new schools could differ.
- PISA 2025 and 2022 values are verified against the OECD report; earlier cycles (2000–2018) come from Wikipedia only.
- All findings are correlational, not causal.

Reproduce: `pip install -r requirements.txt` then `python main.py` (add `--refresh-pisa` to re-scrape Wikipedia).